Michelson interferometer, Haidinger rings & FT spectroscopy

Double-pass arm geometry, dispersive compensator plate C, equal-inclination circular fringes, and doublet visibility beats.

Double-pass virtual air gap & compensator C

Beam splitter BS folds fixed mirror M2 onto virtual plane M2-prime separated by d = |L1 - L2| from movable mirror M1. Translating M1 by dx changes path difference by 2 dx:

$$\Delta_{\text{opt}}(\theta) = 2d\cos\theta = m\lambda \Rightarrow \Delta N = \frac{2\,\Delta x}{\lambda}$$

WHY / Without unsilvered compensator plate C, one arm crosses dispersive BS glass 3 times and the other once — d(lam) = 0 fails across colours and white-light fringes vanish!

WATCH / With identical plate C parallel to BS, one reflection is internal in glass (0) and the other external in air (pi): at exact d = 0 the central white-light fringe is DARK!

Haidinger circular rings vs Fizeau wedge lines

When M1 || M2-prime strictly, extended light produces circular equal-inclination Haidinger fringes localized at infinity; tilting M1 by alpha produces straight wedge fringes!

$$2d\left(1 - \frac{\theta_p^2}{2}\right) = (m_0 - p)\lambda \Rightarrow \theta_p = \sqrt{\frac{p\lambda}{d}}, \qquad \beta_{\text{tilted}} = \frac{\lambda}{2\alpha}$$

WHY / Exactly as in end-on circular YDSE, MAXIMUM order m0 = 2d/lam sits AT CENTRE theta = 0! Shrinking d makes circular rings collapse inward into 0!

WATCH / At d -> 0 one broad fringe fills the entire field ("infinite width"); slight tilt alpha straightens curved arcs into straight lines ONLY right at d = 0!

Gas refractometry & sodium D-doublet beats

Pumping gas into tube length L in one arm traverses the gas TWICE! With a spectral doublet (lam, lam + dlam), incoherent sum of two interferograms beats periodically to zero visibility:

$$n_{\text{gas}} - 1 = \frac{N\lambda_0}{2L}$$

$$V(\Delta x) = \left|\cos\left(\frac{2\pi\,\Delta\lambda\,\Delta x}{\lambda^2}\right)\right| \Rightarrow \Delta x_{\text{beat}} = \frac{\lambda^2}{2\,\Delta\lambda}$$

Sodium D anchor (589.0 & 589.6 nm, dlam = 0.6 nm): mirror travel between consecutive visibility washouts is dx_beat = (589.3e-9)^2 / (2 x 0.6e-9) = 0.289 mm!

WATCH / Watch factor of 2 carefully: mirror travel between washouts is lam^2 / (2 dlam), whereas round-trip optical path difference between washouts is lam^2 / dlam!

Michelson-Morley, LIGO & FTIR resolution

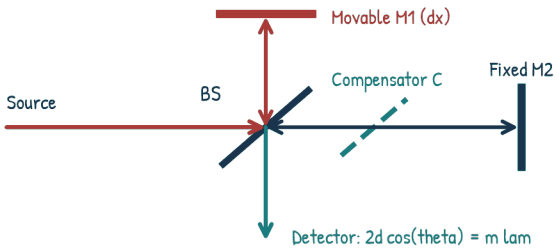
In Fourier-Transform (FTIR) spectroscopy, recording I(Delta) across total optical path scan Delta_max = 2 X_max recovers spectrum S(nu_tilde) with wavenumber resolution 1 / Delta_max:

$$\delta\tilde{\nu}_{\min} = \frac{1}{\Delta_{\max}} = \frac{1}{2X_{\max}} \Rightarrow \mathcal{R}_{\text{FTIR}} = \frac{\tilde{\nu}}{\delta\tilde{\nu}} = \frac{2X_{\max}}{\lambda}$$

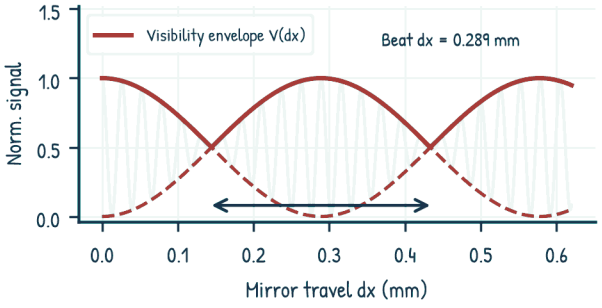
WHY / Michelson-Morley (1887) predicted 2 L v^2 / (lam c^2) ~ 0.4 fringe shift from Earth orbital motion and observed 0; LIGO uses 4 km arms to sense h ~ 10^(-21) strains!

WATCH / A compact 5 cm mirror scan (Delta_max = 10 cm) achieves resolving power R = 200,000 at 500 nm — matching a 200,000-line diffraction grating!

Michelson double-pass arms, virtual air plate d, and glass compensator plate C



Sodium D-doublet interferogram beats vanish every 0.289 mm of mirror travel



REMEMBER / Moving Michelson mirror M1 by dx shifts 2 dx / lam fringes (1 fringe per lam/2); gas cell length L enters TWICE as 2(n - 1)L!

Temporal & spatial coherence, group index & stellar baselines

Wiener-Khinchin Fourier duality, white-light group index shift, van Cittert-Zernike theorem, and Betelgeuse.

Temporal coherence & Wiener-Khinchin theorem

Equal-intensity fringe visibility V(tau) = |gamma(tau)| is the modulus of the Fourier transform of normalised power spectral density S(nu), fixing coherence time tau_c and length L_c:

$$\tau_c \sim \frac{1}{\Delta\nu}, \qquad L_c = c\,\tau_c \approx \frac{\lambda^2}{\Delta\lambda}, \qquad N_{\text{fringes}} \sim \frac{L_c}{\lambda} = \frac{\lambda}{\Delta\lambda}$$

WHY / Top-hat bandwidth gives |sinc(pi dnu tau)| (first zero at tau = 1/dnu); Lorentzian collision broadening gives exp(-pi dnu |tau|); Gaussian Doppler gives Gaussian decay.

WATCH / Sunlight has L_c ~ 1 um (~2 fringes!), Na discharge ~0.5 mm, low-pressure Kr-86 ~50 cm, and single-frequency He-Ne/fibre lasers reach L_c > 100 km!

Dispersion & white-light group-index zero shift!

Insert dispersive plate (n(lam), t) in one arm under WHITE light! Stationary phase d(dphi)/dk0 = 0 places the brightest white-light fringe at GROUP OPL, NOT phase OPL:

$$\Delta_{\text{air}}^{\text{white peak}} = (n_g - 1)\,t, \qquad n_g \equiv \frac{c}{v_g} = n(\lambda) - \lambda\,\frac{dn}{d\lambda}$$

WHY / In normal dispersion (dn/dlam < 0), group index n_g EXCEEDS phase index n! Individual monochrome fringes creep inside the envelope as t varies!

WATCH / IPhO classic trap: crown glass has n = 1.52 while n_g = 1.545 — reading the shifted achromatic peak directly measures n_g - 1 = 0.545 (5% higher than n - 1)!

Spatial coherence width & van Cittert-Zernike

An incoherent source of angular width theta_s = s / D1 washes out two-slit visibility according to the Fourier transform of the source brightness profile:

$$V_{\text{slit source}}(d) = \left|\operatorname{sinc}\!\left(\frac{\pi\, d\,\theta_s}{\lambda}\right)\right| \Rightarrow d_{\text{washout}} = \frac{\lambda}{\theta_s}, \qquad l_{\text{spatial}} \sim \frac{\lambda}{\theta_s}$$

WHY / Even completely independent atoms across a distant star become spatially coherent across baseline l_s ~ lam / theta_* simply by propagating far enough!

WATCH / Solar disc subtends theta_sun = 9.3e-3 rad -> transverse spatial coherence width of raw sunlight on Earth is ONLY ~60 um (why YDSE needs a pinhole first)!

Michelson 20-ft stellar interferometer (Betelgeuse)

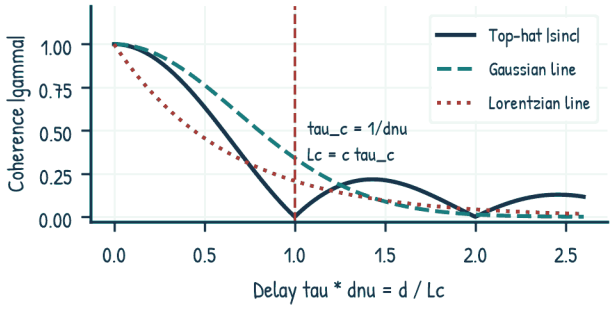
For a uniform circular star disc of angular diameter theta_*, 2D Fourier transform replaces sinc by Airy 2 J1(x)/x, driving fringe visibility to zero at outrigger baseline b:

$$V_{\text{star}}(b) = \left|\frac{2J_1(\pi b\,\theta_\star/\lambda)}{\pi b\,\theta_\star/\lambda}\right| = 0 \Rightarrow \theta_\star = \frac{1.22\,\lambda}{b_{\text{washout}}}$$

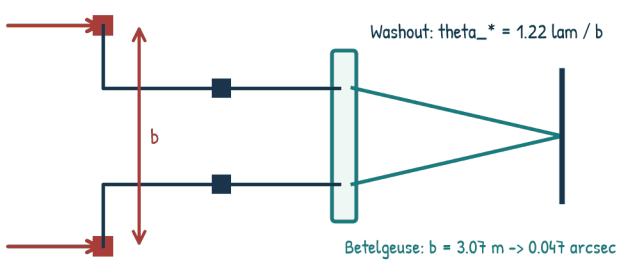
Historic 1920 Betelgeuse benchmark: lam = 575 nm, first fringe disappearance at b = 3.065 m -> theta_* = 2.29e-7 rad = 0.047 arcsec (~400 solar diameters at 640 ly)!

WATCH / Eyepiece fringe SPACING depends on FIXED inner mirror gap d_in (lam f / d_in), whereas fringe CONTRAST depends on outer periscope baseline b!

Wiener-Khinchin transform pairs spectral line shape with fringe visibility



Michelson periscope stellar interferometer beats atmospheric seeing limits



REMEMBER / Dispersive glass shifts the white-light zero fringe by GROUP index (n_g - 1)t, NOT phase index (n - 1)t! Stellar disc washes out at b theta_* = 1.22 lam.

Fresnel amplitude equations, Stokes laws & Bragg stacks

Exact TE (s) & TM (p) field coefficients, power conservation factor, incoherent thick-plate piles, and HR mirrors.

Exact Fresnel amplitude equations (s & p)

Matching tangential E_|| and H_|| across non-magnetic boundary n1 -> n2 (n1 sin i = n2 sin r) gives exact amplitude reflection and transmission ratios:

$$r_s = \frac{n_1\cos i - n_2\cos r}{n_1\cos i + n_2\cos r} = -\frac{\sin(i-r)}{\sin(i+r)}$$

$$r_p = \frac{n_2\cos i - n_1\cos r}{n_2\cos i + n_1\cos r} = \frac{\tan(i-r)}{\tan(i+r)}$$

$$t_s = \frac{2n_1\cos i}{n_1\cos i + n_2\cos r}$$

$$t_p = \frac{2n_1\cos i}{n_2\cos i + n_1\cos r}, \qquad T_{s,p} = \frac{n_2\cos r}{n_1\cos i}|t_{s,p}|^2$$

WATCH / NEVER write T = |t|^2! At air-to-glass normal incidence (n = 1.5), t = 2 / 2.5 = 0.80 -> t^2 = 0.64, whereas TRUE transmitted power is 1.5 x 0.64 = 0.96 (96%)!

Normal incidence, p-sign flip & Stokes relations

At normal incidence R(0) = ((n1 - n2)/(n1 + n2))^2 (4.0% for glass 1.50; 2.0% for water 1.33; 17.2% for diamond 2.42). Reversing rays gives exact Stokes time-reversal laws:

$$r' = -r \quad \text{(time reversal proof of } \pi \text{ flip!)}$$

$$t\,t' + r^2 = 1$$

WHY / Watch r_p closely as i sweeps from 0 to 90 deg on air-glass: r_p crosses ZERO at theta_B = 56.31 deg and flips phase by pi for all grazing angles i > theta_B!

WATCH / As i -> 90 deg (grazing incidence), BOTH R_s -> 100% AND R_p -> 100% with pi phase jump — even rough paper or asphalt acts like a shiny mirror at grazing angle!

Incoherent thick-plate & M-plate pile formulas

For a thick window pane (thickness L >> L_c), multiple internal reflections add INCOHERENTLY as intensities (T0^2 + T0^2 R0^2 + T0^2 R0^4 + ...), NOT amplitudes!

$$T_{1\text{ plate}} = \frac{1-R}{1+R}$$

$$R_{1\text{ plate}} = \frac{2R}{1+R}, \qquad T_{M-\text{plates}} = \frac{1-R}{1+(2M-1)R}$$

Check: single window pane R = 0.04 transmits 0.96 / 1.04 = 92.31% (reflects 7.69%). At theta_B on glass 1.5, R_p = 0 -> T_{M,p} = 100%, while R_s = 25/169 = 14.79%!

WATCH / Distinguish THIN film (L << L_c, coherent sum, oscillates 0 to 4R) from THICK glass window (L >> L_c, incoherent geometric sum, equals 2R/(1+R) smooth)!

Exact quarter-wave AR residual & HR Bragg mirrors

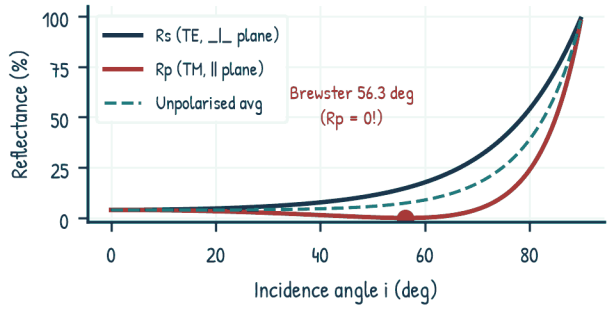
Any quarter-wave dielectric layer (n_c t = lam0 / 4) has exact normal reflectance R_min; alternating N high/low quarter-wave pairs (H/L)^N build 99.99% laser mirrors:

$$R_{\lambda/4} = \left(\frac{n_c^2 - n_1 n_g}{n_c^2 + n_1 n_g}\right)^2, \qquad R_{\text{Bragg},\,N} = \left[\frac{1-(n_g/n_1)(n_H/n_L)^{2N}}{1+(n_g/n_1)(n_H/n_L)^{2N}}\right]^2$$

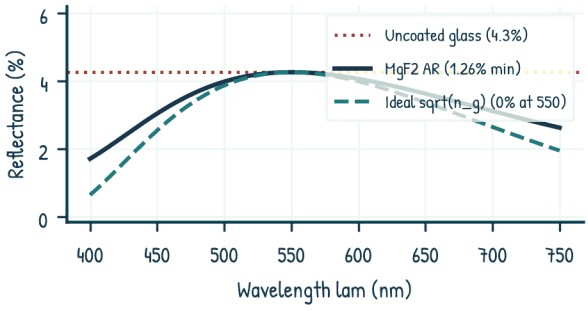
WHY / Single high-index TiO2 quarter-wave layer (n_c = 2.35 > glass 1.52) boosts R UP to 32.3% (lossless beam splitter!), whereas 10 TiO2/SiO2 pairs reach R = 99.98%!

WATCH / Dielectric Bragg mirrors suffer ZERO metallic ohmic heating, letting kilowatt industrial and LIGO cavities handle megawatts of circulating optical power!

Exact air-glass (n = 1.50) Fresnel power reflectances Rs(i) and Rp(i)



Quarter-wave AR spectral curve (550 nm zero explains purple lens tint)



REMEMBER / Reflectance is R = |r|^2, whereas power transmittance carries impedance-projection factor T = (n2 cos r / n1 cos i) |t|^2 = 1 - R!

TIR phase jumps, Fresnel rhomb & evanescent tunnelling

Complex cos(r) beyond critical angle, achromatic Fresnel rhomb QWP, skin decay audit, and frustrated TIR.

TIR unit-modulus reflection & continuous phase

For n1 > n2 (relative n = n2/n1 = sin theta_c < 1) and i > theta_c, cos(r) = i sqrt(sin^2 i / n^2 - 1) makes r_s = exp(-i delta_s) and r_p = exp(-i delta_p) pure phase factors (R = 100%):

$$\tan\left(\frac{\delta_s}{2}\right) = \frac{\sqrt{\sin^2 i - n^2}}{\cos i}$$

$$\tan\left(\frac{\delta_p}{2}\right) = \frac{\sqrt{\sin^2 i - n^2}}{n^2 \cos i}, \qquad \tan\left(\frac{\Delta}{2}\right) = \frac{\cos i \sqrt{\sin^2 i - n^2}}{\sin^2 i}$$

WHY / At i = theta_c both delta_s = delta_p = 0; at i = 90 deg both reach pi; in between, p leads s by peak retardation tan(Delta_max/2) = (1 - n^2)/(2n)!

WATCH / Linearly polarised light striking a TIR prism at azimuth != 0, 90 deg emerges ELLIPTICALLY polarised because delta_p != delta_s!

Fresnel rhomb — achromatic quarter-wave prism

In crown glass (n1 = 1.51 -> n = 0.662), maximum single-bounce retardation is Delta_max = 45.6 deg, hitting exact 45.0 deg at internal angle i = 54.6 deg!

$$\Delta_{2\ \text{bounces}} = 2 \times 45^\circ = 90^\circ = \frac{\pi}{2}$$

$$\Rightarrow 45\ \text{deg linear input} \ \rightarrow\ 100\%\ \text{circular output!}$$

WHY / Unlike crystalline quartz QWP whose phase scales strongly as 1/lam0, Fresnel rhomb depends ONLY on refractive index n(lam), staying ~90 deg across ALL visible colours!

WATCH / Submerging an un-silvered Fresnel rhomb into water raises n = 1.33/1.51 = 0.88, collapsing Delta_max to 7 deg and destroying circular conversion completely!

Evanescent wave & decay-length audit fix (eq. 7.4)

Inside rarer medium 2 (z > 0), field E2 ~ exp(-kappa z) exp(i(k_x x - wt)) skims along x while decaying exponentially along z with ZERO net normal Poynting flux (<S_z> = 0):

$$\kappa = \frac{2\pi}{\lambda_0}\sqrt{n_1^2 \sin^2 i - n_2^2}$$

$$d_E = \frac{1}{\kappa}\ (|E| \propto e^{-z/d_E}), \qquad d_I = \frac{1}{2\kappa}\ (I \propto e^{-2\kappa z} = e^{-z/d_I})$$

WHY / Audit fix (Wave-optics.md eq. 7.4 & Q5): kappa uses 2 pi / lam0 (vacuum lam0!), so intensity/transmission decays as exp(-2 kappa g) = exp(-g / d_I) with 4 pi inside d_I!

WATCH / NEVER put 4 pi inside d AND simultaneously write exp(-2 g / d) — pick ONE notation (exp(-2 kappa g) with kappa = 2 pi/... OR exp(-g / d_I) with d_I = lam0 / 4 pi...)!

Frustrated TIR tunnelling & Goos-Hanchen shift

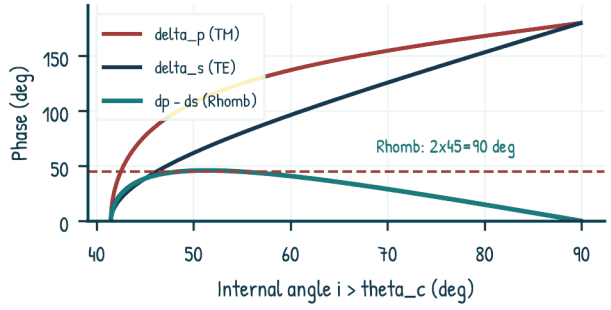
Bringing second glass prism n1 within sub-wavelength gap g (~lam0) couples the evanescent tail back into a propagating wave — exact optical twin of quantum alpha-tunnelling!

$$T_{\text{FTIR}}(g) = \frac{1}{1 + C_{s,p}\sinh^2(\kappa g)} \approx \frac{4}{C_{s,p}} e^{-2\kappa g} \quad (\kappa g \gg 1)$$

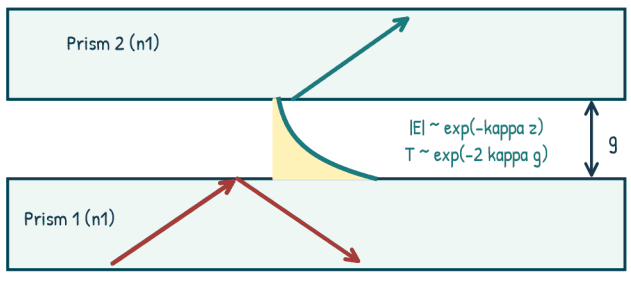
WHY / Touching skin ridges to a TIR prism frustrates air gap g -> 0 along ridges while valleys keep g >> d_E, imaging fingerprints sharply; TIRF microscopy illuminates only 80 nm of cells!

WATCH / A narrow laser beam undergoing TIR also steps laterally forward along the boundary by Goos-Hanchen shift dx_GH = -d(delta)/dk_x ~ O(lam0)!

TIR phase shifts in crown glass: two 45-deg bounces make an achromatic QWP



Evanescent field skin depth and Frustrated TIR optical barrier tunnelling



REMEMBER / Keep amplitude decay length d_E = 1/kappa SEPARATE from intensity length d_I = 1/(2 kappa) so factor 2 is NEVER double-counted!

Exact N-slit polygon, blazed echelle & acousto-optic gratings

Exact (sin N gamma / sin gamma)^2 factor, subsidiary peaks, oblique Littrow gratings, and ultrasonic phase gratings.

Exact N-slit phasor sum & subsidiary peaks

Summing N equal slits of width a and spacing d gives geometric series sum_{q=0}^{N-1} exp(i 2 q gamma) = exp(i(N-1)qamma) sin(N qamma)/sin(qamma):

$$I(\theta) = I_1(0)\left(\frac{\sin(N\gamma)}{\sin\gamma}\right)^2\left(\frac{\sin\beta}{\beta}\right)^2 \qquad \left(\gamma = \frac{\pi d \sin\theta}{\lambda},\quad \beta = \frac{\pi a \sin\theta}{\lambda}\right)$$

WHY / Principal peaks hit N^2 I1 with half-width dgamma = pi/N (~1/N width), conserving total energy N^2 x (1/N) = N; first side lobe sits at ~1/(3 pi / 2)^2 = 4.5% of peak!

WATCH / Count carefully: between consecutive principal maxima lie N - 1 dark zeros (N gamma = p pi, p not multiple of N) and N - 2 weak subsidiary maxima!

Oblique grating equation, minimum deviation & Littrow

With incident ray at angle i and diffracted ray at theta_m on the SAME side of normal (reflection convention), path difference across qroove step d is d(sin i + sin theta_m):

$$d(\sin i + \sin\theta_m) = m\lambda$$

$$m_{\max}^{\text{oblique}} = \left\lfloor \frac{2d}{\lambda} \right\rfloor, \qquad 2d\sin\left(\frac{D_{\min}}{2}\right) = m\lambda$$

WHY / In auto-collimating Littrow mount (diffracted ray returns straight back along incident ray, theta_m = i!), 2 d sin(i) = m lam gives R = mN = 2 W sin(i) / lam!

WATCH / Normal incidence caps orders at floor(d/lam), whereas grazing incidence + grazing diffraction doubles reach to floor(2d/lam)!

Free spectral range & sawtooth blazed echelles

Unblazed gratings dump most energy into useless zero order m = 0. Tilting reflective groove facets by blaze angle theta_b steers specular reflection straight onto target order m!

$$\Delta\lambda_{\text{FSR}} = \frac{\lambda}{m}, \qquad \lambda_{\text{blaze}}^{\text{Littrow}} = \frac{2d\sin\theta_b}{m}$$

WHY / Astronomical echelle spectrographs run steep facets (theta_b ~ 63.4 deg) at orders m = 50-100 for R > 100,000, using a cross-disperser prism to separate stacked orders!

WATCH / Higher order m sharpens resolution (R = mN) while squeezing Free Spectral Range (lam / m) — exact same tradeoff as Fabry-Perot etalons!

2D meshes, optical discs & ultrasonic gratings

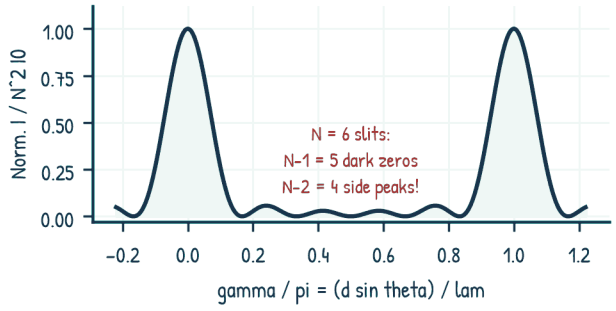
Crossed 2D mesh (d_x, d_y) forms reciprocal grid spots (p lam / d_x, q lam / d_y). In water, an ultrasonic wave of speed v_s and frequency f_s creates refractive index ripples:

$$d_{\text{acoustic}} = \lambda_{\text{sound}} = \frac{v_s}{f_s} \Rightarrow \lambda_{\text{sound}}\sin\theta_m = m\lambda_{\text{light}}$$

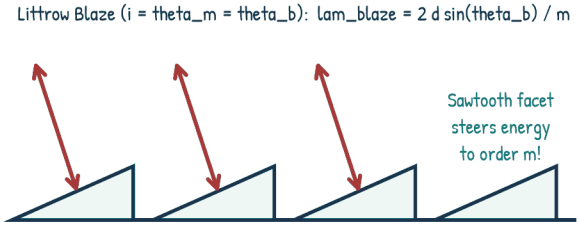
WHY / Debye-Sears acousto-optic modulators steer laser beams electronically in microseconds simply by tuning RF drive frequency f_s!

WATCH / Instantaneous refractive-index grating period equals FULL acoustic wavelength lam_sound = v_s / f_s (NOT lam_sound / 2)!

Exact N-slit interference factor sharpening principal lines as width ~ 1/N



Sawtooth blazed reflection echelle grating concentrates 85% power into order m



REMEMBER / Between any two adjacent principal grating peaks sit STRICTLY N - 1 dark zeros and N - 2 subsidiary peaks; maximum oblique order is floor(2d / lam)!

Fabry-Perot etalons, Airy finesse & laser cavities

Infinite multiple-beam geometric series, 100% resonant transmission miracle, finesse vs F, and Wiener standing wave.

Exact Airy formula & 100% resonance miracle

Two parallel lossless mirrors of reflectance R = |r|^2 spaced by n L sum geometric series T0 / (1 - R exp(i delta)) with round-trip phase delta = 4 pi n L cos(theta) / lam0:

$$T_{\mathrm{FP}}(\delta) = \frac{1}{1 + F\sin^2(\delta/2)}$$

$$R_{\mathrm{FP}}(\delta) = \frac{F\sin^2(\delta/2)}{1 + F\sin^2(\delta/2)}, \qquad F \equiv \frac{4R}{(1-R)^2}$$

WHY / Even with TWO 99% mirrors, on resonance (delta = 2m pi) leakage from giant intracavity power 1/(1 - R) cancels front reflection destructively, giving T_FP = 100%!

WATCH / Mirror absorption A > 0 (R + T0 + A = 1) spoils cancellation and caps peak transmission at T_max = [T0 / (T0 + A)]^2!

Coefficient F vs Finesse F_cal (notation alert!)

Resonances repeat every 2 pi in phase while each Lorentzian-like spike has full width at half maximum ddelta_FWHM = 4 / sqrt(F) = 2(1 - R) / sqrt(R):

$$\mathcal{F} \equiv \frac{2\pi}{\delta_{\mathrm{FWHM}}} = \frac{\pi\sqrt{R}}{1-R} = \frac{\pi\sqrt{F}}{2}$$

$$(R = 0.96 \Rightarrow \mathcal{F} = 77.0, \quad F = 3840)$$

WHY / Finesse F_cal counts how many round-trip beams interfere constructively inside the cavity — exact multiple-beam analogue of grating slit count N!

WATCH / NEVER plug F = 4R/(1 - R)^2 into R = m F_cal! Chromatic resolving power uses FINESSE pi sqrt(R)/(1 - R), which is smaller by ~sqrt(F)!

Free spectral range & hyperfine resolving power

Normal resonance m lam = 2 n L separates longitudinal modes by Free Spectral Range c / (2 n L) and achieves ultra-high chromatic resolving power m * F_cal:

$$\Delta\nu_{\mathrm{FSR}} = \frac{c}{2nL}$$

$$\Delta\lambda_{\mathrm{FSR}} = \frac{\lambda^2}{2nL}, \qquad \mathcal{R} = \frac{\lambda}{\delta\lambda_{\min}} = m\,\mathcal{F} = \frac{2nL}{\lambda}\,\frac{\pi\sqrt{R}}{1-R}$$

Anchor: L = 1 cm air etalon, R = 0.97 (F_cal = 103), lam = 500 nm -> order m = 40,000 -> R = 4.1e6 (resolves 0.12 pm Zeeman/hyperfine splittings)!

WATCH / He-Ne laser (gain width 1.5 GHz, cavity L = 30 cm -> FSR = 0.50 GHz) oscillates on ~3 simultaneous longitudinal modes spaced by 500 MHz!

Wiener 1890 standing light wave experiment

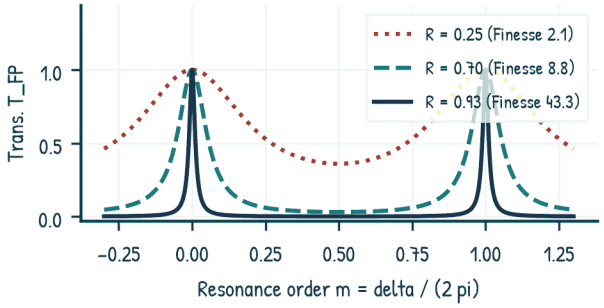
Normal silver-mirror reflection r = -1 forms optical standing wave |E(z)| = 2 E0 |sin(kz)| with E-nodes at m lam/2 and B-antinodes shifted by lam/4 (B maximum AT z = 0!):

$$z_{E-\mathrm{antinode}} = \left(m + \tfrac{1}{2}\right)\tfrac{\lambda}{2}, \qquad D_{\mathrm{emulsion\ bands}} = \frac{\lambda}{2\sin\alpha}$$

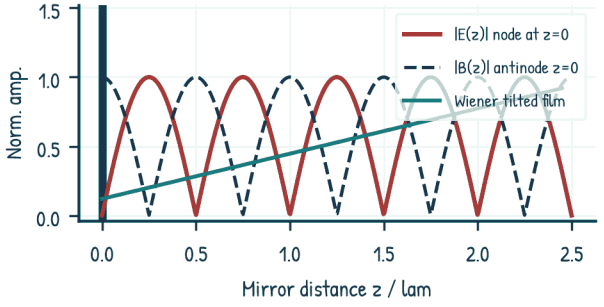
WHY / Wiener sliced the standing field with a 20 nm film tilted by alpha ~ 1e-3 rad: zero blackening at mirror z = 0 proved silver grains respond to ELECTRIC vector E, not B!

WATCH / In fiber-optic Bragg sensors and Lippmann colour plates, standing waves write refractive layers spaced by lam / (2 n), reflecting that exact wavelength back!

Fabry-Perot Airy resonances reach T = 100% even when each mirror reflects 99%



Wiener 1890 experiment: emulsion blackens at E antinodes, proving E acts on matter



REMEMBER / Distinguish coefficient F = 4R/(1 - R)^2 from reflective finesse F_cal = pi sqrt(R)/(1 - R); resolving power equals R = m * F_cal!

Fresnel half-period zones, zone plates & Cornu spiral

Equal-area concentric zones, A1/2 spiral convergence, Arago bright spot, Babinet principle, and straight-edge I0/4.

Fresnel half-period zones & 4 I0 aperture miracle

Divide plane wavefront at distance b from axial point P into rings whose outer radii r_m add path m lam / 2: EVERY zone has equal area pi lam b and opposite sign!

$$r_m = \sqrt{m\lambda b} \quad \left(\text{or } \sqrt{\tfrac{m\lambda ab}{a+b}}\right)$$

$$A_{\text{open}} = A_1 - A_2 + A_3 - \ \ldots = \tfrac{A_1}{2} \Rightarrow I_{1\ \text{zone}} = 4I_0$$

WHY / Opening ONLY zone 1 blocks zone 2 destructive cancellation, making axial spot P FOUR TIMES brighter than removing the entire opaque screen!

WATCH / Widening hole to include 2 zones gives A1 - A2 ~ 0 (DARKNESS on axis!); odd number of zones -> ~4 I0 bright, even number -> ~0 dark!

Arago-Poisson bright spot & Babinet principle

A smooth coin/ball blocking the first k zones leaves A_{k+1} - A_{k+2} + ... = A_{k+1} / 2 ~ A1 / 2 -> UNOBSTRUCTED brightness I0 right at the dead centre of its shadow!

$$E_{\text{aperture}}(P) + E_{\text{obstacle}}(P) = E_{\text{open}}(P)$$

$$\Rightarrow I_{\text{wire}}(\theta \neq 0) = I_{\text{slit}}(\theta \neq 0)$$

WHY / Babinet principle equates diffraction away from theta = 0 for ANY complementary aperture/obstacle pair — how laser micrometers measure human hair thickness w via w sin(theta) = m lam!

WATCH / Arago spot requires a smooth circular rim and spatially coherent illumination — rough jagged coin edges scramble zone phases and erase the central spot!

Fresnel zone plate — flat multi-focus X-ray lens

Blocking all even zones lets all M open odd zones add IN PHASE at primary focal length f1 = r1^2 / lam AND subsidiary odd foci f1/3, f1/5, ... (plus diverging virtual foci -f_k!):

$$f_k = \frac{r_1^2}{(2k-1)\lambda}$$

$$I_{\text{focus, amp}} = 4M^2 I_0, \qquad I_{\text{focus, phase plate}} = 16M^2 I_0$$

WHY / Etching even zones to phase depth t = lam0 / [2(n - 1)] flips -A_{2m} to +A_{2m}, doubling amplitude again andboosting focal intensity 4x over black-ring plates!

WATCH / Zone-plate focal length varies inversely with wavelength (f1 ~ 1/lam: red focuses CLOSER than blue — opposite to refractive glass lenses)!

Cornu clothoid spiral & straight-edge shadow law

Fresnel integrals C(v), S(v) coil between eyes (-1/2, -1/2) and (+1/2, +1/2) (open chord squared = 2 -> I0). A semi-infinite razor edge blocks v < 0:

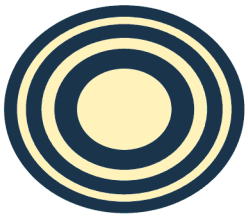
$$I_{\text{geometric edge}}(v=0) = \tfrac{I_0}{2}\left[\left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2\right] = \mathbf{\tfrac{I_0}{4}}$$

$$I_{\text{1st bright}} = 1.370\, I_0$$

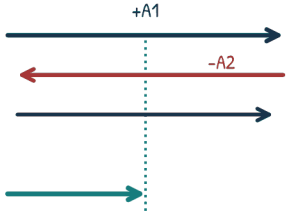
WHY / Cutting half the wavefront halves ELECTRIC FIELD AMPLITUDE to A0 / 2, so irradiance on the geometric shadow line equals (1/2)^2 I0 = I0 / 4 = 25%!

WATCH / Inside the shadow of a single straight edge intensity decays monotonically to zero with ZERO dark fringes; dark fringes appear ONLY on the lit side!

Fresnel half-period zone plate and alternating spiral sum A1 - A2 + A3 - ... = A1/2

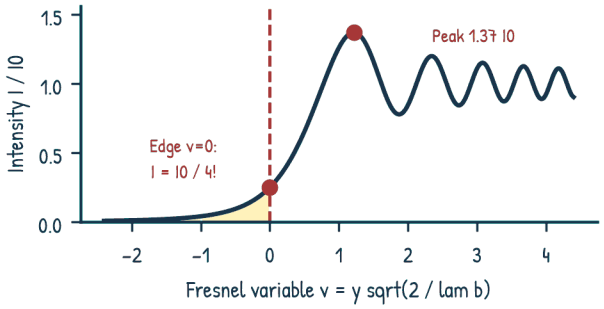


Zone radii r_m = sqrt(m lam b)



A_open = A1/2 => 1-zone I = 4 I0!

Fresnel straight-edge shadow transition: exact I0/4 at geometric edge v = 0



REMEMBER / 1 open Fresnel zone gives 4 I0 (2 zones give ZERO!); right on a straight razor shadow edge intensity is I0 / 4, NEVER I0 / 2!

Abbe Fourier imaging, X-ray Bragg law, Sagnac & HBT

Back-focal-plane spatial filtering, Zernike phase contrast, crystal Bragg glancing law, fiber gyros, and HBT bunching.

Abbe spatial filtering & Zernike phase contrast

Objective lens forms spatial Fourier transform of specimen at back focal plane F-prime (spot spacing x_m = m lam f / Lambda). Image periodicity reconstructs iff >= 2 orders pass:

$$\Lambda_{\min}^{\text{axial coherent}} = \frac{\lambda_0}{\mathrm{NA}}, \qquad \Lambda_{\min}^{\text{oblique cone}} = \frac{\lambda_0}{2\,\mathrm{NA}}$$

WHY / Blocking DC spot m = 0 in F-prime yields Dark-Field imaging; retarding m = 0 by pi/2 converts invisible cell phase shifts phi(x) into direct intensity contrast 1 +/- 2 phi(x)!

WATCH / Putting a vertical slit mask inside F-prime passes horizontal diffraction orders and erases horizontal lines while leaving vertical specimen lines sharp!

3D X-ray Bragg crystal reflection & n < 1 shift

Crystal lattice planes spaced by d_{hkl} = a0 / sqrt(h^2 + k^2 + l^2) reflect X-rays at GLANCING angle theta when adjacent-plane path difference 2 d sin(theta) equals n lam:

$$2d_{hkl}\sin\theta = n\lambda \quad (\lambda \le 2d_{hkl})$$

$$n_{\text{X ray}} = 1 - \delta \Rightarrow 2d\sin\theta_n\left(1 - \frac{\delta}{\sin^2\theta_n}\right) = n\lambda$$

WHY / Because X-ray frequencies sit ABOVE atomic electron resonances, refractive index n = 1 - delta < 1 (delta ~ 10^(-5)), causing external grazing TIR below theta_c = sqrt(2 delta)!

WATCH / NUMBER-ONE X-RAY TRAP: if X-rays hit at 60 deg TO THE SURFACE NORMAL, glancing Bragg angle is theta = 90 - 60 = 30 deg (and beam deflection is 2 theta = 60 deg)!

Sagnac rotating ring & Fresnel-drag cancellation

Counter-propagating beams around rotating loop area A (angular speed Omega, N fiber turns) experience opposite beam-splitter motion, accumulating Sagnac phase shift:

$$\Delta t = \frac{4N\Omega\cdot\mathbf{A}}{c^2}, \qquad \Delta\Phi_{\text{Sagnac}} = \frac{8\pi N\Omega\cdot\mathbf{A}}{\lambda_0 c}, \qquad \Delta\nu_{\text{RLG}} = \frac{4\Omega\cdot\mathbf{A}}{\lambda_0 P}$$

WHY / Relativistic Fresnel drag v = c/n +/- R Omega (1 - 1/n^2) inside co-moving silica fiber cancels n^2 identically, leaving Delta_Phi INDEPENDENT of refractive index n!

WATCH / IPhO/APhO favourite red herring: stem gives n = 1.46 silica fiber and asks for new Sagnac shift vs air loop — answer is UNCHANGED (1 : 1)!

Hanbury Brown-Twiss (HBT) & 3D holography

HBT correlates intensity fluctuations <I1(t) I2(t + tau)> electrically between separated telescopes; thermal photon bunching locks g^(2)(tau) directly to |gamma_12|^2:

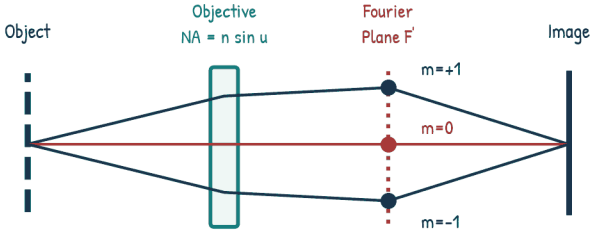
$$g^{(2)}(\tau) = \frac{\langle I_1(t)\, I_2(t+\tau)\rangle}{\langle I_1\rangle\langle I_2\rangle} = 1 + |\gamma_{12}(\tau)|^2$$

$$I_{\text{holo}} = |E_r + E_o|^2 = I_r + I_o + E_r^*\,E_o + E_r E_o^*$$

WHY / Phase turbulence cancels out of |gamma_12|^2 in HBT; in off-axis holography, interference fringes encode BOTH amplitude and phase of E_o onto flat film!

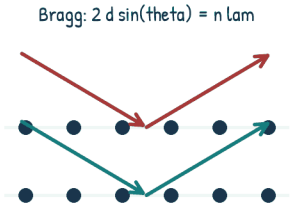
WATCH / Cutting a hologram into quarters does NOT chop the 3D image into quarters — EVERY piece still reconstructs the ENTIRE 3D object with slightly lower resolution!

Abbe two-step Fourier imaging and spatial filtering in back focal plane F-prime



Need >= 2 orders inside aperture => d_min = lam0 / (2 NA)

Crystal X-ray Bragg reflection (glancing angle!) and rotating Sagnac interferometer



theta measured FROM plane!



dPhi = 8 pi Omega.A / (lam0 c)

REMEMBER / Bragg angle theta is ALWAYS measured from the crystal plane (NOT normal!); fiber Sagnac phase shift is 100% independent of core index n!